

Choosing the Right LLM Deployment Strategy for Spring AI

Practice 3 Transcript | Oct 1, 2026 15:13

Dr. Aria Collins 00:00

Thanks for joining. Our leadership team is firm on one point—no sensitive company data can leave our walls. With that constraint in mind, what's your proposed strategy for running our Spring AI application?

You 01:12

We will configure our Spring AI application to route all prompts exclusively to self-hosted, local LLMs like Ollama running on our internal infrastructure, guaranteeing zero data exfiltration by eliminating external API calls entirely.

Dr. Aria Collins 01:14

That sounds like a solid start! Keeping everything internal really aligns with the no-data-leaving mandate. Now, have you considered how you'll manage the performance demands of hosting those LLMs locally, or how you'll handle updates to the models?

You 03:50

Performance demands and model updates will be managed by deploying a load-balanced inference cluster running high-throughput engines like vLLM for quantized models, alongside an air-gapped internal registry that enables secure, zero-downtime rollouts without external internet access.

Dr. Aria Collins 03:52

Really smart approach! Load balancing with vLLM and an air-gapped registry checks a lot of boxes for scalability and security. How do you plan to handle encryption for data in transit and at rest within this setup?

You 04:29

Data in transit will be secured using TLS 1.3 with mutual authentication (mTLS) across all internal microservice and GPU node communications, while data at rest will be protected using AES-256 encryption across all vector databases, model registries, host file systems, and temporary caches.

Dr. Aria Collins 04:32

Absolutely spot on! TLS 1.3 and AES-256 are top-of-the-line choices for securing sensitive data. One last thing—how are you planning to address compliance requirements, like audit trails or data access controls, in this deployment?

You 04:55

Compliance will be maintained by integrating Spring Security with our internal identity provider for strict role-based access controls, alongside an immutable, centralized logging system that records all user identities, prompt payloads, and system actions to guarantee comprehensive audit trails.

Dr. Aria Collins 04:57

That's really comprehensive! Role-based access and immutable logs are fantastic measures for both compliance and accountability. I think your strategy checks all the right boxes—privacy, performance, and compliance. Great work putting this together!

End of session